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MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : HU-611

PRODUCTION & OPERATIONAL MANAGEMENT

Time Allotted: 3 Hours | *Full Marks*: 70

The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable.

Group – A

(Multiple Choice Type Questions)

1. Choose the correct alternatives for the followings (*any ten*): 1×10=10

(i) Which one of the following is not a decision taken during the aggregate production planning stage?

(a) Scheduling of machine
(b) amount of labour to be committed
(c) Rate at which production should happen
(d) Inventory to be carried forward

(ii) Which of the following factors are to be carried for production scheduling?

(1) Sales forecast (2) Component design (3) Rate sheet (4) Time standard

Select the correct answer using the codes given below codes.

(a) 1,2 and 3
(b) 1,2 and 4
(c) 1,3 and 4
(d) 2,3 and 4

(iii) Each product requires a purchased component A is its sub assembly. Before the start of week-1, there are 400 component of type A in stock. The lead time to procure this component is 2 weeks and the order quality is 400. Number of components A per product is only one. The manufacturer should place the order for

(a) 400 components in week-1
(b) 400 components inweek-3
(c) 200 components in week-1 and 200 component in week-3
(d) 400 components in week-5

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(iv) Which one of the following charts gives simultaneously information about the progress of work and machine loading?

(a) Process chart
(b) Machine load chart
(c) Man-machine chart
(d) Gantt Chart

(v) Classifying item in A,B, and C categories for selective control in inventory management is done by arriving items in the decreasing order of

(a) Total inventory cost
(b) Item value
(c) Annual usage value
(d) Item demand

(vi) In the ABC method of inventory control, Group A constitutes costly items. What is the percentage of such item of the total items?

(a) 10 to 20%
(b) 20 to 30%
(c) 30 to 40%
(d) 40 to 50%

(vii) In the basic EOQ model, if lead time increases from 5 to 10 days, the EOQ will:

(a) Double
(b) Decrease by a factor of two
(c) Remain same
(d) Insufficient data

(viii) Economic order quantity is the quantity at which the cost of carrying is

(a) Minimum
(b) Equal to the cost of ordering
(c) Less than the cost of ordering
(d) Cost of over-stock

(ix) If the annual demand of an item becomes half, ordering cost double, holding cost one-fourth and the unit cost twice, then what is the ratio of the new EOQ and the earlier EOQ?

(a) $\frac{1}{2}$
(b) $\sqrt[1]{2}$
(c) $\sqrt{2}$
(d) 2

(x) Each product requires a purchased component A in its sub assembly. Before the start of week-1 there are 400 components of type A in stock. The lead time to procure this component is 2 weeks and the order for :

(a) 400 components in week-1
(b) 400 components in week-3
(c) 200 components in week -1 and 200 components in week-3
(d) 400 components in week-5

Group – B

(Short Answer Type Questions)

Answer *any three* of the following. 3×5=15

2. 'Production is one of the main function in achieving organisation objectives'- Justify. 5
3. Explain Q- System and P- System of inventory control. 5

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4. Fit a straight line trend by least square method to the following figures of production of a sugar factory

Year	2001	2002	2003	2004	2005	2006	2007
Production In,000tons	76	85	95	81	91	96	90

Estimate the production for 2008. 5

5. What is control Chart? Why it is used? What are the advantages of control chart? 5

6. Mention the various components of lean manufacturing. 5

Group – C

(Long Answer Type Questions)

Answer *any three* of the following. 3×15=45

7. (a) What do you mean by Quality control? Differentiate between inspection and Quality control. 7+8

(b) A manufacturer of LED television has a final inspection before the product is packed for shipping. The final inspection involves testing the television on several parameters and verifying if the set performs satisfying in all the tests. Periodically, 10 television sets are drawn for testing and the following table shows the performance during one such test.

Sample No.	1	2	3	4	5	6	7	8	9	10
Number of Defects	11	9	12	9	12	22	7	10	13	6

Draw a control chart for number of defects and comment whether the process is under control.

8. (a) What do you mean by Crashing of project network?

(b) A construction project involves the following activities with their precedence relation and respective time duration.

Activity	A	B	C	D	E	F	G	H
Immediate Predecessor	-	-	A	B,C	B	E	D,F	E
Duration(Days)	3	16	5	8	4	4	17	11

(i) Draw a network diagram, indicate the critical path, and calculate the project duration.
(ii) Calculate EST, LST, EFT, LFT for each activity. 3+7+5

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9. (i) What is material management? What are its weeks, scopes and advantages? 5

(ii) In a study of its sales, a motor company obtained the following least square trend equation; Y=1600+200×: (origin 1950, × units = 1 Year, Y =total no. of units sold annually) The company has physical facilities to produce only 3600 units a year and it believes that at least for the next decade the trend will continue as before.

(a) What is the average annual increase in the number of units sold?
(b) By what year the company's expected sales have equalled its present physical capacity?
(c) Estimate the annual sales for 1965. How much in excess of the company's present physical capacity is this estimated value? 10

10. (i) The demand rate for a particular item is 12000 units/ Year. The ordering cost is Rs.100 Per order and the holding cost is Rs. 0.80 per item per month. If no shortage are allowed and the replacement is instantaneous, determine:

(1) The economic order quantity.
(2) The time between orders.
(3) The number of orders per year.
(4) The optimum annual cost if the cost of item is Rs.2 per item.

(ii) Explain the logic behind ABC Analysis. 10+5

11. (a) There are five jobs, which are to be processed on two machines A and B in the order AB. The processing times in hours for the jobs are given below. Find the optimal sequence and total elapsed time. Also find the idle times of the two machines.

JOBS	1	2	3	4	5
MACHINE A (TIME IN hrs)	2	6	4	8	10
MACHINE B (TIME IN hrs)	3	1	5	9	17

(b) A TV repairman finds that the time spend on this jobs have as exponential distribution which means of 30 minutes. If the repairs set is the order in which they come in and if the arrival of sets is approximately. Poisson with an average rate of 10 per 8 hrs. day. What is repairman's expected idle time each day? How many jobs are ahead of the average set just brought is? 9+6

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